[Total No. of Questions - 9] [Total No. of Printed Pages - 2]

Dec-25-0362

CSPC-511 (Computer Networks)

B.Tech. 5th (CBCS/NEP) [CSE/CSE (AI&DS)/CSE (AI&ML)]

Time : 3 Hours **Max. Marks : 60**

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all, i.e one question each from sections A, B, C, D and the compulsory question from section E.

SECTION - A

1. Discuss the goals and applications of Computer Networks. (12)

2. How is TCP/IP reference model different from OSI reference model? Explain in detail. (12)

SECTION - B

3. Explain the following concepts:

   (a) Framing

   (b) Error control (2×6=12)

4. Compare the following protocols:

   (a) Sliding window and Simplex protocol

   (b) Protocol using selective repeat & Protocol using go back N. (2×6=12)

SECTION - C

5. What are the design issues of Network layer? Explain them in detail. (12)

6. Describe the working of Transport Layer Protocol TCP and UDP. (12)

SECTION - D

7. How are data compression techniques useful? Explain their working mechanism. (12)

8. How application layer carries out file transfer, access and management? Explain in detail. (12)

SECTION - E (Compulsory)

9. (a) Discuss the concept of FDM.

   (b) How is LAN different from MAN?

   (c) List the services provided by data link layer to network layer.

   (d) Why congestion needs to be controlled?

   (e) Enlist the use of virtual terminal.

   (f) Why is optimal routing employed? (6×2=12)